

MatMul — C = A × B on the T1 fabric (3×3 worked example)

Steps run top → bottom. Colour helps tracking here. The V-load transpose turns columns into B^T rows can be seen with colour; each Output cell keeps its final-position colour, so a result enters at column 0 and shifts right into place.

Example

0. Reference
C = A × B (true result)

1	2	3
4	5	6
7	8	9

B	10	11	12
	13	14	15
	16	17	18

84	90	96
201	216	231
318	342	366

Expected

Setup

1. Load H / V
vle8 v4,(A) [H]
csrwi 0x7c0,1
vle8 v5,(B) [V] → B

1	2	3
4	5	6
7	8	9

B ^T	10	13	16
	11	14	17
	12	15	18

Iteration 0 (Full Steps)

2. k=2 Broadcast
csrwi 0x7c0,1
vrgather.vx
v6,v5,2 [V]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	15	18
12	15	18
12	15	18

3. k=2 Multiply
csrwi 0x7c0,0
vmul.vv v6,v4,v6 [H]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	30	54
48	75	108
84	120	162

4. k=2 Reduce
vredsum.vs
v7,v6,v28 →
v7[0]=C[*][2]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	30	54	Σ	96		
48	75	108	Σ	231		
84	120	162	Σ	366		

5. Shift Right
vslideup.vi
v8,v7,1
vmv.v.v v7,v8

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

->	96		
->	231		
->	366		

SAME AS ABOVE 6. Broadcast, 7. k=1 Multiply

shift one column right →

Iteration 1 (Simplified)

8. k=1 Reduce
vredsum.vs
v7,v6,v28 →
v7[0]=C[*][1]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

11	28	51	Σ	90	96	
44	70	102	Σ	216	231	
77	112	153	Σ	342	366	

9. Shift Right
vslideup.vi
v8,v7,1
vmv.v.v v7,v8

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

->	90	96
->	216	231
->	342	366

SAME AS ABOVE 10. Broadcast, 11. k=0 Multiply

shift one column right →

Iteration 2 (Simp)

12. k=0 Reduce
vredsum.vs
v7,v6,v28 → full C
vse8 v7,(C)

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

10	26	48	Σ	84	90	96
40	65	96	Σ	201	216	231
70	104	144	Σ	318	342	366

Answer!!